MIL-T-82389(TD)
15 October 1969
SUPERSEDING
3313-421A
15 September 1968

MILITARY SPECIFICATION
TRAINING AID, SLEEVE BEARING

This specification has been approved by the Naval Training Device Center, Department of the Navy

1. SCOPE

1.1 This specification covers one type of an enlarged scale mockup of a sleeve bearing, to serve as a classroom demonstrator.

1.1.1 This item has been designated by the Naval Training Device Center as Device 29FA8 for Navy Procurement (see 6.2)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-601	Boxes Wood Cleated-plywood
PPP-B-621	Boxes, Wood, Nailed and Lockcorner

Military

MIL-P-116	Preservation; Methods of
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-L-10547	Liners, Case and Sheet, Overwrap, Water-vaporproof or Waterproof, Flexible
MIL-T-23991	Training Devices, Military; General Specification for

FSC 6910

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Naval Training Device Center (NAVTRADEVCEN)

3313-63-401	Drawing for Device 29FA8

PHOTOGRAPHS

Naval Training Device Center (NAVTRADEVCEN)

NP(40)3264-1 through NP(40)3264-5	Photographs of Device 29FA8

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Materials.- Materials used shall be in accordance with 3.1.1 through 3.1.5. Materials not specifically described shall be of the best quality, of the lightest practicable weight, suitable for the purpose intended and selected with consideration given to the requirements of 3.3 through 3.5.

3.1.1 Metals.- Metals used shall be either corrosion-resistant or shall be suitably treated to resist corrosion under environmental conditions specified in 3.3.

3.1.2 Bridge gage.- The bridge gage shall be aluminum or brass.

3.1.3 Plywood.- Plywood, when used, shall have a minimum of five (5) plies or be solid core. Outer surfaces shall be hardwood and all plies or solid core shall be free of voids.

3.1.4 Wood.- Wood used for upper and lower bearing case and pedestal shall be select grade white poplar having good appearance and finishing qualities, suitable for either clear or opaque finish. Wood shall be properly seasoned, free from checks, splits, cross gain or seasoning defects which may cause warping.

3.1.4.1 Glue.- Only plastic resin glue shall be used. Any glue adversely affected by moisture or lacquer shall not be used.

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3.1.5 Plastic.- Plastic used for babbitt liner, shall be epoxy, polyvinyl chloride or a material of equal shock and abrasion resistance to endure without visible imperfection or deterioration. Colors shall be molded in plastic and not painted on.

3.2 Design.- The training aid shall consist of a tabletop mockup, constructed to demonstrate the details of sleeve bearing construction and methods of measuring bearing clearance. The design shall be in accordance with the drawings of NAVTRADEVCEN 3313-63-401.

3.3 Reliability.- The training aid shall withstand continuous classroom use of 12 hours per week for 5 years, and not be adversely affected in any manner during storage and transportation when under ambient conditions of temperature ranging from -5 to 125 degrees Fahrenheit and of relative humidity ranging up to 95 percent. The training aid shall be assembled in such a manner that no fixed part, assembly or control shall become undesirably free or sluggish in operation, and no degradation shall be caused in the mechanical performance stipulated for the training aid because of shipment, operation or storage.

3.4 Construction.- The device shall be constructed in accordance with the process requirements of MIL-T-23991.

3.4.1 Breakage and deformation.- Construction shall be such as to minimize the possibility of breakage and deformation during use by the instructor or students.

3.4.2 Tolerances.- All tolerances, not otherwise specified, shall be consistent with functional design, inherent qualities of the material being used and shall afford freedom from binding in mechanical fits.

3.5 Performance.- The training aid shall be such that the instructor will be able to demonstrate the function, assembly and disassembly of component parts and methods of measuring bearing clearances.

3.6 Details of components.- The training aid shall consist of the following major components. Component arrangement shall be as shown in the photographs of NAVTRADEVCEN NP(40)3264-1 through NP(40)3264-5.

(a) Sleeve bearing model

(b) Base

(c) Dust cover

(d) Accessories.

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3.6.1 Sleeve bearing model.- The sleeve bearing model shall meet the design requirements of 3.2.

3.6.2 Base.- The base of the training aid shall meet the materials requirements of 3.1.3.

3.6.3 Dust cover.- A dust cover shall be fabricated of 16 ounce, light gray vinyl-covered nylon to cover the training aid, including the base, when not in use. Covering material used shall be compatible with all finishes of the training aid so that no migrating solvents, contained within the material, will adversely affect any part of the training aid.

3.6.4 Accessories.- The training aid shall include the following accessories:

(a) Bridge gage

(b) Micrometer depth gage - Lufkin No. 513 or equal in range accuracy, durability and method of operation

(c) Thickness (feeler) gage - Lufkin No. 126 or equal in accuracy, durability and quantity of leaves.

3.6.4.1 Bridge gage.- The bridge gage shall have adjusting screws for varying tolerances.

3.7 Finish.- All surfaces to be painted shall be sealed, primed and then finished with a satin-surface lacquer in accordance with the Preparation for Painting and Painting requirement of MIL-T-23991. Application of finishes shall be accomplished in such a manner as not to impair performance of the equipment.

3.7.1 All aluminum parts not painted shall be anodized in accordance with MIL-A-8625.

3.7.2 All finished surfaces shall be free from tool marks, burrs, scratches, sharp edges or blemishes.

3.8 Color.- Colors for the training aid shall be as specified by the Contracting Officer.

3.9 Nameplates or product markings.- A Government-furnished nameplate and emblem will be furnished by the procuring agency.

3.9.1 Nameplate and emblem.- The nameplate and emblem shall be affixed to the training aid in the locations specified by the procuring agency.

3.10 Nomenclature plates.- Nomenclature plates shall be provided and located in a prominent position on the base when viewed from the front. These plates shall be of laminated plastic, or aluminum, and shall be permanently affixed to the base of the training aid.

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3.10.1 Markings.- Markings on nomenclature plates shall be engraved or photochemically indicated and shall be in strong contrast with the background material. A non-reflective surface shall be used.

3.11 Government-furnished property (GFP).- The nameplate and emblem of 3.9 will be furnished by the procuring agency.

3.12 Government-loaned property.- One set of color samples will be loaned for use in the fabrication of the training aid.

3.13 Workmanship.- The training aid shall be constructed and finished with particular attention given to neatness, correction of detail, marking and identification of parts and freedom from sharp edges and burrs.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspections to be performed are classified as follows:

(a) In process inspection

(b) Quality conformance inspection.

4.3 In-process inspection.- In-process inspection shall include such visual and mechanical examinations and testing of materials, subassemblies, parts and accessories (including purchased standard parts) during the manufacturing process as may be necessary to assure conformance to all the requirements of this specification.

4.4 Quality conformance inspection.- The quality conformance inspection shall consist of a preproduction inspection and an inspection of the production units.

4.4.1 A preproduction inspection shall be conducted on a preproduction unit and shall consist of all the examinations and tests specified in 4.5 and 4.6.

4.4.2 Production units shall be inspected to the extent necessary to maintain quality control and conformance to the Government accepted preproduction sample.

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4.5 Examinations.- Examinations shall be performed to verify conformance to the following requirements:

(a) Materials

(b) Construction

(c) Details of components

(d) Tolerances

(e) Finish and color

(f) Nameplates or product markings

(g) Workmanship

(h) Preparation for delivery.

4.6 Tests.- Each training aid shall be subjected to a functional test consisting of disassembly and reassembly to determine if the training aid conforms to the requirements and design of 3.1 through 3.2.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be Level A or C as specified by the procuring agency.

5.1.1 Level A.- For level A, each device shall be preserved and packaged for mechanical and physical protection in accordance with MIL-P-116, Method III.

5.1.2 Level C.- For Level C, preservation and packaging shall conform to the manufacturer's commercial practice.

5.2 Packing.- Packing shall be Level A or C as specified by the procuring agency.

5.2.1 Level A.- Units packaged in accordance with 5.1.1 shall be packed in exterior containers conforming to PPP-B-621, Class 2 or PPP-B-601, Styles A or B with steel banding. The gross weight shall not exceed the weight limitations of the shipping container. Each exterior shipping container shall be lined with a waterproof moisture barrier sealed with adhesive or tape to effect water resistance in accordance with MIL-L-10547. A case liner shall not be required where the package(s) dunnage have been treated to resist water.

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5.2.2 Level C.- Units packaged as specified in 5.1.2 shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with common carrier rules and regulations as applicable to the mode of transportation and as may be the supplier's commercial practice when such meets the requirements of this level.

5.3 Physical protection.- All material shall be packaged and packed in containers constructed to provide a compact and nonshifting load. Adequate cushioning, blocking and bracing shall be provided within shipping containers in a manner designed to prevent damage due to shock, vibration or impact during shipment, handling and storage.

5.4 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The training aid is intended for use to familiarize students in the assembly, disassembly and procedure in measuring bearing clearances.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Procuring Activity designation (see 1.1.1)

(c) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

CUSTODIAN:
Navy-TD

PREPARING ACTIVITY:
Navy-TD
Project No. 6910-N201

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specificacion in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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